SECTION 5: MONEY AND BANKING (II)

5.1 MONETARY BASE (M0) AND MONETARY POLICY

MARKING SCHEME

1994/AL/II/18 A	2005/AL/II/19 C (55%)	1990/AL/II/03 A	2000/AL/II/08 B	2012/DSE/I/30 A (51%)
1995/AL/II/16 C	2005/AL/II/20 D (56%)	1990/AL/II/04 C	2003/AL/II/12 D	2013/DSE/I/28 B (43%)
1995/AL/II/17 D	2007/AL/II/7 A (30%)	1991/AL/II/23 B	2004/AL/II/04 B (74%)	2017/DSE/I/32 A (61%)
1995/AL/II/28 C	2011/AL/II/20 C (66%)	1992/AL/II/04 B	2005/AL/II/17 B (77%)	2017/DSE/I/37 B (66%)
1996/AL/II/19 D	2014/DSE/I/33 A (70%)	1993/AL/II/16 A	2006/AL/II/07 A (84%)	2019/DSE/I/41 B
1999/AL/II/15 C	2015/DSE/I/29 B (43%)	1996/AL/II/18 B	2007/AL/II/10 A (63%)	2020/DSE/I/30 B
2001/AL/II/13 D	2015/DSE/I/32 C (64%)	1997/AL/II/08 A	2010/AL/II/18 B (73%)	2020/DSE/I/32 A
2002/AL/II/21 C (29%)	2018/DSE/I/32 C (40%)	1998/AL/II/17 A	2012/AL/II/14 A (64%)	2021/DSE/I/34 B
2021/DSE/I/35 A	2023/DSE/32 C	2024/DSE/II/28 B	2024/DSE/II/35 D	2024/DSE/II/39 A

1994/AL/II/3

(a) When the government purchases bonds from the market, it increases the reserves of commercial banks, the lending ability of the latter (through the banking multiplier) and hence the money supply.

(b) With the reserve ratio at 0.2, 1 million dollars of additional reserves can back up a loan of 5 millions. Thus the money supply increases by an amount of 5 million.

In the calculation, we have assumed that banks are fully loaned up and there is no cash leakage.

1996/AL/II/5(a)

As the public buy government bonds, the reserves in the commercial banks are reduced. Assuming that the banks are fully loaned up, their credit creation power is reduced and less loan can be made. The money supply falls.

2004/AL/II/1(c)

S: Monetary aggregates, both including currency (banknotes and coins) held by the non-bank public.

D: M0 (currency + reserves) vs. M1 / M2 / M3 (currency + deposits).

2007/AL/II/3(a)

The monetary base (or high-powered money) is the sum of currency held by the non-bank public (C) and reserves (R, deposits of commercial banks at the central bank plus vault cash).

The money supply is the sum of currency in circulation (C) and bank deposits (D).

Money supply is often a multiple of base money because, under fractional reserve banking, commercial banks are only required to keep a fraction of their deposits as reserves and are thus able to use their excess reserves to generate further deposits through their loan and portfolio investment activities.

The banking multiplier equals unity when commercial banks keep 100% reserves against their deposits.

2008/AL/II/1(c)
S: interest rates charged on short-term loans to commercial banks;
D: rate set by the central bank at the discount window **vs.** rate set by the (overnight) interbank loans market.

2012/DSE/II/13
(a) Required reserve ratio = ($500 million - $100 million) / $2 000 million (1)
= 20% (1)

(b) Maximum possible amount of deposits = $500 million × (1 / 0.2) (1)
= $2 500 million (1)

(c) Deposit = $700 million × (1 / 0.2) = $3 500 million (2)
Cash in public circulation = 0 (1)
Money supply = Deposit + Cash in public circulation = $3 500 million (1)

2013/DSE/II/12
(a) Monetary base = $400 million + $1 000 million = $1 400 million (2)
Money supply = $1 000 million + $2 000 million = $3 000 million (2)

(b) New money supply = $1 000 million + [$4 000 × (1 / 25%)] = $2 600 million (2)

2014/DSE/II/7(b)
Advantages: An increase in money supply would lower the interest rate and raise investment, leading to an increase in real output in the short run (given upward-sloping aggregate supply). Given idle resources in the economy, more labour (and/or capital) would be used to produce the bigger output, implying an increase in employment or decrease in unemployment. (Contractionary fiscal policy would not result in an increase in GDP and employment.) (2)

Disadvantages: In the long run (with vertical aggregate supply), an increase in money supply would not create any effect on output and employment. But if the fiscal deficits grow so fast that the money supply has to be increased at a higher rate that of real output, high inflation would even arise. (Contractionary fiscal policy would not result in inflation.) (2)

2014/DSE/II/12
(a) Monetary base = cash in public circulation + reserves held by commercial banks (1)
The monetary base increases because the commercial banks have more reserves. (2)

(b) Credit creation process:
When the central bank purchases bonds from commercial banks, more cash is injected to the commercial banks as reserves and there would be excess reserves in the banking system. (1)
The banks would lend out the excess reserves. (1)
And the bank loans will be re-deposited into the banking system. (1)
[The process will go on and on (until the actual reserves are equal to required reserves).]

No. The monetary base will remain unchanged as the sum of cash in public circulation and reserves in the banking system will not be affected by the credit creation process. (2)

2015/DSE/II/6
(a) (i) Legal reserve ratio = $1 000 million / $5 000 million = 0.2 (1)

(ii) Money supply = $150 million + $5 000 million = $5 150 million (2)

(b) New monetary base = $150 million + $1 050 million = $1 200 million (2)

(c) New money supply = $150 million + [$1 050 million × (1 / 0.2)] = $5 400 million (3)

2016/DSE/II/8

(a) Monetary base = \$500 million + \$1 000 million = \$1 500 million (1)
Money supply = \$500 million + \$4 000 million = \$4 500 million (1)

(b) Change in monetary base = \$1 500 million - \$1 500 million = 0 (2)

Change in money supply
= New money supply - Original money supply
= [\$500 million + \$1 000 million × (1 / 50%)] - \$4 500 million = -\$2 000 million (2)

OR
= Change in cash held by non-bank public + Change in deposits
= 0 + [\$1 000 million × (1 / 50%) - \$4 000 million) = -\$2 000 million (2)

(c) (i) Under a fractional reserve banking system, banks are required to keep only a fraction of their deposits as reserves. They can create credit by lending out the remaining fraction in the form of loans to businesses and private individuals, which would eventually find their way back into the banking system in the form of deposits, resulting in a multiple increase in deposits / a banking multiplier larger than unity. (3)

(ii) If the legal reserve ratio is 100% (RRR = 1), the money supply will be equal to the monetary base.
OR
If all of the banks decide not to lend out their excess reserves even when RRR < 1 (e.g. during financial crisis, when default rates are high), the money supply will be equal to the monetary base. (1)

2018/DSE/II/7(a)

(i) New M_0 = \$500 million + \$100 million + \$40 million = \$640 million (2)

(ii) Banks do not hold excess reserves / there is sufficient demand for loans. (1)
There is no cash leakage. (1)

(iii) Old M_S = \$2 000 million + \$100 million = \$2 100 million
RRR = (\$500 million - \$100 million) / \$2 000 million = 0.2 (1)
New M_S = \$100 million + [(\$500 million + \$40 million) × (1 / 0.2)] = \$2 800 million (2)
∴ M_S increases by \$700 million (= \$2 800 million - \$2 100 million). (1)

[Credits would also be awarded to those candidates who reasoned that the public's initial cash holding of \$100 million might also be deposited in to the banking system to yield \$500 million of additional deposits, resulting in an aggregate increase of \$1 200 million of deposits and thus a net increase of \$1 100 million of money supply. As a side note, for the topic of money and banking at the DSE level, cash held by the general public should be treated as fixed (i.e., change of cash in public is assumed to be zero except under other changes).]

2023/DSE/II/12c

(c) Increase in discount rate raises the cost for commercial banks to borrow money from the Federal Reserve. Monetary base tends to decrease. **(1)**
Money supply decreases. **(1)**
Interest rates increase.[1] **(1)**
Investment decreases. **(1)**
Aggregate demand decreases. Price level tends to decrease, relieving the pressure on the rising price level. **(1)**

2025/DSE/II/10(Cross Topic 5.1 /5.2 /5.3 /6.1)

Marks

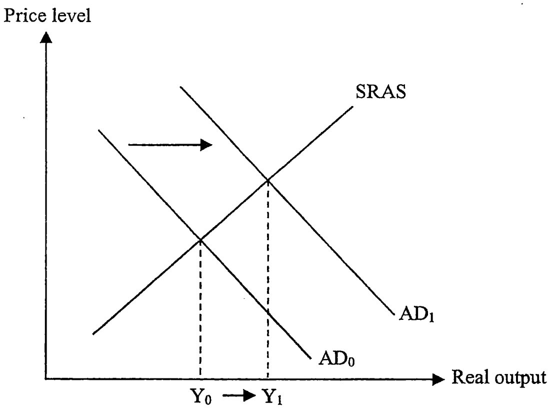
(a) (i) Money supply growth rate = (20 837.4 – 21 691.3) / 21 691.3 x 100% = -3.94% (1)

(ii) QTM: MV = PY, where M is money supply, V is velocity of circulation of money, P is general price level, Y is real output, assuming V to be constant. (2)
Growth rate of Y ≈ growth rate of M – growth rate of P = -3.94 – 3.6 = -7.54% (1)

(b) As discount rate is lowered, the cost for commercial banks to borrow money from the Federal Reserve decreases. (1)
Money supply increases. (1)
Nominal interest rate decreases. (1)

(c) Verbal elaboration:
- As interest rate decreases, investment increases. (1)
- Aggregate demand increases. (1)
- Real output increases. (1)

Indicate in the diagram:
- rightward shift of AD curve (1)
- increase in Y (1)



(d) Argument FOR:
- Increasing expenditure on infrastructure can increase the capital endowment in the future, raising the aggregate output in the long run. (2)

Argument AGAINST:
- Increasing expenditure on infrastructure would raise the US Government's expenses, worsening the fiscal balance. (2)